Task 1:

```
bodike@bodike-chaithali:~/Home$ mkdir B24CS013_Chaithali
bodike@bodike-chaithali:~/Home$ ls
B24CS013_Chaithali
bodike@bodike-chaithali:~/Home$ cd B24CS013_Chaithali/
bodike@bodike-chaithali:~/Home/B24CS013_Chaithali$ mkdir dir1 dir2 dir3 dir4 dir5
bodike@bodike-chaithali:~/Home/B24CS013_Chaithali$ ls
dir1  dir2  dir3  dir4  dir5
bodike@bodike-chaithali:~/Home/B24CS013_Chaithali$ cd dir1
bodike@bodike-chaithali:~/Home/B24CS013_Chaithali/dir1$ ls
bodike@bodike-chaithali:~/Home/B24CS013_Chaithali/dir1$ touch file1.txt file2.txt file3.txt file4.txt file5.py
bodike@bodike-chaithali:~/Home/B24CS013_Chaithali/dir1$ ls
file1.txt  file2.txt  file3.txt  file4.txt  file5.py
bodike@bodike-chaithali:~/Home/B24CS013_Chaithali/dir1$
```

Q1)

```
bodike@bodike-chaithali:~/Home$ mkdir B24CS013_Chaithali
bodike@bodike-chaithali:~/Home$ ls
B24CS013_Chaithali
bodike@bodike-chaithali:~/Home$ cd B24CS013_Chaithali/
bodike@bodike-chaithali:~/Home/B24CS013_Chaithali$ mkdir dir1 dir2 dir3 dir4 dir5
bodike@bodike-chaithali:~/Home/B24CS013_Chaithali$ ls
dir1  dir2  dir3  dir4  dir5
bodike@bodike-chaithali:~/Home/B24CS013_Chaithali$ cd dir1
bodike@bodike-chaithali:~/Home/B24CS013_Chaithali/dir1$ ls
bodike@bodike-chaithali:~/Home/B24CS013_Chaithali/dir1$ touch file1.txt file2.txt file3.txt file4.txt file5.py
bodike@bodike-chaithali:~/Home/B24CS013_Chaithali/dir1$ ls
file1.txt  file2.txt  file3.txt  file4.txt  file5.py
bodike@bodike-chaithali:~/Home/B24CS013_Chaithali/dir1$ cp file1.txt ../dir4
bodike@bodike-chaithali:~/Home/B24CS013_Chaithali/dir1$ cd ..
bodike@bodike-chaithali:~/Home/B24CS013_Chaithali$ cd dir4
bodike@bodike-chaithali:~/Home/B24CS013_Chaithali/dir4$ ls
file1.txt
bodike@bodike-chaithali:~/Home/B24CS013_Chaithali/dir4$ cd ..
bodike@bodike-chaithali:~/Home/B24CS013_Chaithali$ cd dir1
bodike@bodike-chaithali:~/Home/B24CS013_Chaithali/dir1$ mv ../dir4/file1.txt ../dir4/file6.txt
bodike@bodike-chaithali:~/Home/B24CS013_Chaithali/dir1$ cd ..
bodike@bodike-chaithali:~/Home/B24CS013_Chaithali$ ls dir4
file6.txt
bodike@bodike-chaithali:~/Home/B24CS013_Chaithali$ chmod 654 ../dir4/file6.txt
chmod: cannot access '../dir4/file6.txt': No such file or directory
bodike@bodike-chaithali:~/Home/B24CS013_Chaithali$ cd dir1
bodike@bodike-chaithali:~/Home/B24CS013_Chaithali/dir1$ chmod 654 ../dir4/file6.txt
bodike@bodike-chaithali:~/Home/B24CS013_Chaithali/dir1$ cd ..
bodike@bodike-chaithali:~/Home/B24CS013_Chaithali$ ls -l dir4/file6.txt
-rw-r-xr-- 1 bodike bodike 0 Aug  2 16:15 dir4/file6.txt
bodike@bodike-chaithali:~/Home/B24CS013_Chaithali$
```

Q2)

```
bodike@bodike-chaithali:~/Home$ ls
B24CS013_Chaithali
bodike@bodike-chaithali:~/Home$ cd B24CS013_Chaithali/
bodike@bodike-chaithali:~/Home/B24CS013_Chaithali$ mkdir dir1 dir2 dir3 dir4 dir5
bodike@bodike-chaithali:~/Home/B24CS013_Chaithali$ ls
dir1  dir2  dir3  dir4  dir5
bodike@bodike-chaithali:~/Home/B24CS013_Chaithali$ cd dir1
bodike@bodike-chaithali:~/Home/B24CS013_Chaithali/dir1$ ls
bodike@bodike-chaithali:~/Home/B24CS013_Chaithali/dir1$ touch file1.txt file2.txt file3.txt file4.txt file5.py
bodike@bodike-chaithali:~/Home/B24CS013_Chaithali/dir1$ ls
file1.txt  file2.txt  file3.txt  file4.txt  file5.py
bodike@bodike-chaithali:~/Home/B24CS013_Chaithali/dir1$ cp file1.txt ../dir4
bodike@bodike-chaithali:~/Home/B24CS013_Chaithali/dir1$ cd ..
bodike@bodike-chaithali:~/Home/B24CS013_Chaithali$ cd dir4
bodike@bodike-chaithali:~/Home/B24CS013_Chaithali/dir4$ ls
file1.txt
bodike@bodike-chaithali:~/Home/B24CS013_Chaithali/dir4$ cd ..
bodike@bodike-chaithali:~/Home/B24CS013_Chaithali$ cd dir1
bodike@bodike-chaithali:~/Home/B24CS013_Chaithali/dir1$ mv ../dir4/file1.txt ../dir4/file6.txt
bodike@bodike-chaithali:~/Home/B24CS013_Chaithali/dir1$ cd ..
bodike@bodike-chaithali:~/Home/B24CS013_Chaithali$ ls dir4
file6.txt
bodike@bodike-chaithali:~/Home/B24CS013_Chaithali$ chmod 654 ../dir4/file6.txt
chmod: cannot access '../dir4/file6.txt': No such file or directory
bodike@bodike-chaithali:~/Home/B24CS013_Chaithali$ cd dir1
bodike@bodike-chaithali:~/Home/B24CS013_Chaithali/dir1$ chmod 654 ../dir4/file6.txt
bodike@bodike-chaithali:~/Home/B24CS013_Chaithali/dir1$ cd ..
bodike@bodike-chaithali:~/Home/B24CS013_Chaithali$ ls -l dir4/file6.txt
-rw-r-xr-- 1 bodike bodike 0 Aug  2 16:15 dir4/file6.txt
bodike@bodike-chaithali:~/Home/B24CS013_Chaithali$ cd dir5
bodike@bodike-chaithali:~/Home/B24CS013_Chaithali/dir5$ ls
bodike@bodike-chaithali:~/Home/B24CS013_Chaithali/dir5$ mv ../dir1/file3.txt .
bodike@bodike-chaithali:~/Home/B24CS013_Chaithali/dir5$ ls
file3.txt
bodike@bodike-chaithali:~/Home/B24CS013_Chaithali/dir5$ chmod 777 file3.txt
bodike@bodike-chaithali:~/Home/B24CS013_Chaithali/dir5$ ls -l file3.txt
-rwxrwxrwx 1 bodike bodike 0 Aug  2 16:12 file3.txt
bodike@bodike-chaithali:~/Home/B24CS013_Chaithali/dir5$
```

Q3)

```
bodike@bodike-chaithali:~/Home/B24CS013_Chaithali/dir1$ touch file1.txt file2.txt file3.txt file4.txt file5.py
bodike@bodike-chaithali:~/Home/B24CS013_Chaithali/dir1$ ls
file1.txt  file2.txt  file3.txt  file4.txt  file5.py
bodike@bodike-chaithali:~/Home/B24CS013_Chaithali/dir1$ cp file1.txt ../dir4
bodike@bodike-chaithali:~/Home/B24CS013_Chaithali/dir1$ cd ..
bodike@bodike-chaithali:~/Home/B24CS013_Chaithali$ cd dir4
bodike@bodike-chaithali:~/Home/B24CS013_Chaithali/dir4$ ls
file1.txt
bodike@bodike-chaithali:~/Home/B24CS013_Chaithali/dir4$ cd ..
bodike@bodike-chaithali:~/Home/B24CS013_Chaithali$ cd dir1
bodike@bodike-chaithali:~/Home/B24CS013_Chaithali/dir1$ mv ../dir4/file1.txt ../dir4/file6.txt
bodike@bodike-chaithali:~/Home/B24CS013_Chaithali/dir1$ cd ..
bodike@bodike-chaithali:~/Home/B24CS013_Chaithali$ ls dir4
file6.txt
bodike@bodike-chaithali:~/Home/B24CS013_Chaithali$ chmod 654 ../dir4/file6.txt
chmod: cannot access '../dir4/file6.txt': No such file or directory
bodike@bodike-chaithali:~/Home/B24CS013_Chaithali$ cd dir1
bodike@bodike-chaithali:~/Home/B24CS013_Chaithali/dir1$ chmod 654 ../dir4/file6.txt
bodike@bodike-chaithali:~/Home/B24CS013_Chaithali/dir1$ cd ..
bodike@bodike-chaithali:~/Home/B24CS013_Chaithali$ ls -l dir4/file6.txt
-rw-r-xr-- 1 bodike bodike 0 Aug  2 16:15 dir4/file6.txt
bodike@bodike-chaithali:~/Home/B24CS013_Chaithali$ cd dir5
bodike@bodike-chaithali:~/Home/B24CS013_Chaithali/dir5$ ls
bodike@bodike-chaithali:~/Home/B24CS013_Chaithali/dir5$ mv ../dir1/file3.txt .
bodike@bodike-chaithali:~/Home/B24CS013_Chaithali/dir5$ ls
file3.txt
bodike@bodike-chaithali:~/Home/B24CS013_Chaithali/dir5$ chmod 777 file3.txt
bodike@bodike-chaithali:~/Home/B24CS013_Chaithali/dir5$ ls -l file3.txt
-rwxrwxrwx 1 bodike bodike 0 Aug  2 16:12 file3.txt
bodike@bodike-chaithali:~/Home/B24CS013_Chaithali/dir5$ cp file3.txt copy_file3.txt
bodike@bodike-chaithali:~/Home/B24CS013_Chaithali/dir5$ ls
copy_file3.txt  file3.txt
bodike@bodike-chaithali:~/Home/B24CS013_Chaithali/dir5$ ls -l copy_file3.txt
-rwxrwxr-x 1 bodike bodike 0 Aug  2 16:45 copy_file3.txt
bodike@bodike-chaithali:~/Home/B24CS013_Chaithali/dir5$ chmod u-w,g-w copy_file3.txt
bodike@bodike-chaithali:~/Home/B24CS013_Chaithali/dir5$ ls -l copy_file3.txt
-r-xr-xr-x 1 bodike bodike 0 Aug  2 16:45 copy_file3.txt
bodike@bodike-chaithali:~/Home/B24CS013_Chaithali/dir5$
```

Q4)

```
bodike@bodike-chaithali:~/Home/B24CS013_Chaithali/dir5$ sudo adduser chaithali_02
Adding user `chaithali_02' ...
Adding new group `chaithali_02' (1000) ...
Adding new user `chaithali_02' (1000) with group `chaithali_02' ...
Creating home directory `/home/chaithali_02' ...
Copying files from `/etc/skel' ...
New password:
BAD PASSWORD: The password fails the dictionary check - it is based on a dictionary word
Retype new password:
passwd: password updated successfully
Changing the user information for chaithali_02
Enter the new value, or press ENTER for the default
        Full Name []: Chaithali
        Room Number []: B-306
        Work Phone []: 123456789
        Home Phone []: 123456789
        Other []: NA
Is the information correct? [Y/n] y
bodike@bodike-chaithali:~/Home/B24CS013_Chaithali/dir5$ sudo usermod -aG sudo chaithali_02
bodike@bodike-chaithali:~/Home/B24CS013_Chaithali/dir5$ sudo su
root@bodike-chaithali:/home/bodike/Home/B24CS013_Chaithali/dir5# cd /home/chaithali_02
root@bodike-chaithali:/home/chaithali_02# ls
root@bodike-chaithali:/home/chaithali_02# cp -r /home/bodike/Desktop/Sem3/CSP203/Practice/B24CS013_Chaithali /home/chaithali_02/
root@bodike-chaithali:/home/chaithali_02# ls
B24CS013_Chaithali
root@bodike-chaithali:/home/chaithali_02#
```

Q5)

```
bodike@bodike-chaithali:~/Home/B24CS013_Chaithali/dir5$ sudo adduser chaithali_02
Adding user `chaithali_02' ...
Adding new group `chaithali_02' (1000) ...
Adding new user `chaithali_02' (1000) with group `chaithali_02' ...
Creating home directory `/home/chaithali_02' ...
Copying files from `/etc/skel' ...
New password:
BAD PASSWORD: The password fails the dictionary check - it is based on a dictionary word
Retype new password:
passwd: password updated successfully
Changing the user information for chaithali_02
Enter the new value, or press ENTER for the default
        Full Name []: Chaithali
        Room Number []: B-306
        Work Phone []: 123456789
        Home Phone []: 123456789
        Other []: NA
Is the information correct? [Y/n] y
bodike@bodike-chaithali:~/Home/B24CS013_Chaithali/dir5$ sudo usermod -aG sudo chaithali_02
bodike@bodike-chaithali:~/Home/B24CS013_Chaithali/dir5$ sudo su
root@bodike-chaithali:/home/bodike/Home/B24CS013_Chaithali/dir5# cd /home/chaithali_02
root@bodike-chaithali:/home/chaithali_02# ls
root@bodike-chaithali:/home/chaithali_02# cp -r /home/bodike/Desktop/Sem3/CSP203/Practice/B24CS013_Chaithali /home/chaithali_02/
root@bodike-chaithali:/home/chaithali_02# ls
B24CS013_Chaithali
root@bodike-chaithali:/home/chaithali_02# groups chaithali_02
chaithali_02 : chaithali_02 sudo
root@bodike-chaithali:/home/chaithali_02# sudo deluser chaithali_02 sudo
Removing user `chaithali_02' from group `sudo' ...
Done.
root@bodike-chaithali:/home/chaithali_02# groups chaithali_02
chaithali_02 : chaithali_02
root@bodike-chaithali:/home/chaithali_02#
```

Q6)

```
New password:
BAD PASSWORD: The password fails the dictionary check - it is based on a dictionary word
Retype new password:
passwd: password updated successfully
Changing the user information for chaithali_02
Enter the new value, or press ENTER for the default
        Full Name []: Chaithali
        Room Number []: B-306
        Work Phone []: 123456789
        Home Phone []: 123456789
        Other []: NA
Is the information correct? [Y/n] y
bodike@bodike-chaithali:~/Home/B24CS013_Chaithali/dir5$ sudo usermod -aG sudo chaithali_02
bodike@bodike-chaithali:~/Home/B24CS013_Chaithali/dir5$ sudo su
root@bodike-chaithali:/home/bodike/Home/B24CS013_Chaithali/dir5# cd /home/chaithali_02
root@bodike-chaithali:/home/chaithali_02# ls
root@bodike-chaithali:/home/chaithali_02# cp -r /home/bodike/Desktop/Sem3/CSP203/Practice/B24CS013_Chaithali /home/chaithali_02/
root@bodike-chaithali:/home/chaithali_02# ls
B24CS013_Chaithali
root@bodike-chaithali:/home/chaithali_02# groups chaithali_02
chaithali_02 : chaithali_02 sudo
root@bodike-chaithali:/home/chaithali_02# sudo deluser chaithali_02 sudo
Removing user `chaithali_02' from group `sudo' ...
Done.
root@bodike-chaithali:/home/chaithali_02# groups chaithali_02
chaithali_02 : chaithali_02
root@bodike-chaithali:/home/chaithali_02# exit
exit
bodike@bodike-chaithali:~/Home/B24CS013_Chaithali/dir5$ sudo deluser --remove-home chaithali_02
[sudo] password for bodike:
Looking for files to backup/remove ...
Removing files ...
Removing user `chaithali_02' ...
Warning: group `chaithali_02' has no more members.
Done.
bodike@bodike-chaithali:~/Home/B24CS013_Chaithali/dir5$
```

Q7)

```
bodike@bodike-chaithali:~/Home/B24CS013_Chaithali/dir5$ cd ../dir3
bodike@bodike-chaithali:~/Home/B24CS013_Chaithali/dir3$ ls
bodike@bodike-chaithali:~/Home/B24CS013_Chaithali/dir3$ cp ../dir4/file6.txt .
bodike@bodike-chaithali:~/Home/B24CS013_Chaithali/dir3$ ls
file6.txt
bodike@bodike-chaithali:~/Home/B24CS013_Chaithali/dir3$ mv file6.txt file7.txt
bodike@bodike-chaithali:~/Home/B24CS013_Chaithali/dir3$ ls
file7.txt
bodike@bodike-chaithali:~/Home/B24CS013_Chaithali/dir3$ cd ..
bodike@bodike-chaithali:~/Home/B24CS013_Chaithali$ cd ..
bodike@bodike-chaithali:~/Home$ ls
B24CS013_Chaithali
bodike@bodike-chaithali:~/Home$ rm -R B24CS013_Chaithali/
rm: remove write-protected regular empty file 'B24CS013_Chaithali/dir5/copy_file3.txt'? y
bodike@bodike-chaithali:~/Home$ ls
bodike@bodike-chaithali:~/Home$
```

Task 2:

Q1)

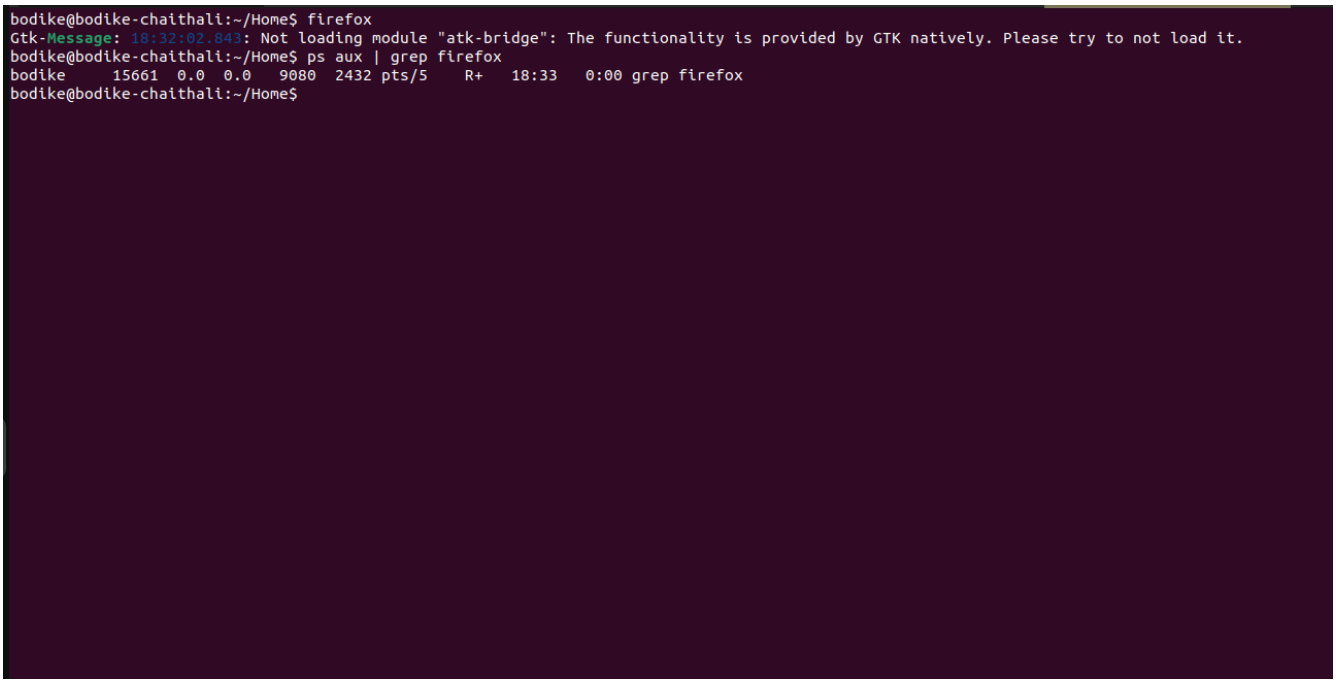
```
bodike@bodike-chaithali:~/Home$ ps aux
USER         PID %CPU %MEM    VSZ   RSS TTY      STAT START   TIME COMMAND
root           1  0.0  0.1 167108 11572 ?        Ss   14:05   0:02 /sbin/init splash
root           2  0.0  0.0      0     0 ?        S    14:05   0:00 [kthreadd]
root           3  0.0  0.0      0     0 ?        S    14:05   0:00 [pool_workqueue_release]
root           4  0.0  0.0      0     0 ?        I<   14:05   0:00 [kworker/R-rcu_g]
root           5  0.0  0.0      0     0 ?        I<   14:05   0:00 [kworker/R-rcu_p]
root           6  0.0  0.0      0     0 ?        I<   14:05   0:00 [kworker/R-slub_]
root           7  0.0  0.0      0     0 ?        I<   14:05   0:00 [kworker/R-netns]
root           9  0.0  0.0      0     0 ?        I<   14:05   0:00 [kworker/0:0H-events_highpri]
root          12  0.0  0.0      0     0 ?        I<   14:05   0:00 [kworker/R-mm_pe]
root          13  0.0  0.0      0     0 ?        I    14:05   0:00 [rcu_tasks_kthread]
root          14  0.0  0.0      0     0 ?        I    14:05   0:00 [rcu_tasks_rude_kthread]
root          15  0.0  0.0      0     0 ?        I    14:05   0:00 [rcu_tasks_trace_kthread]
root          16  0.0  0.0      0     0 ?        S    14:05   0:00 [ksoftirqd/0]
root          17  0.0  0.0      0     0 ?        I    14:05   0:07 [rcu_preempt]
root          18  0.0  0.0      0     0 ?        S    14:05   0:00 [migration/0]
root          19  0.0  0.0      0     0 ?        S    14:05   0:00 [idle_inject/0]
root          20  0.0  0.0      0     0 ?        S    14:05   0:00 [cpuhp/0]
root          21  0.0  0.0      0     0 ?        S    14:05   0:00 [cpuhp/2]
root          22  0.0  0.0      0     0 ?        S    14:05   0:00 [idle_inject/2]
root          23  0.0  0.0      0     0 ?        S    14:05   0:00 [migration/2]
root          24  0.0  0.0      0     0 ?        S    14:05   0:00 [ksoftirqd/2]
root          26  0.0  0.0      0     0 ?        I<   14:05   0:00 [kworker/2:0H-events_highpri]
root          27  0.0  0.0      0     0 ?        S    14:05   0:00 [cpuhp/4]
root          28  0.0  0.0      0     0 ?        S    14:05   0:00 [idle_inject/4]
root          29  0.0  0.0      0     0 ?        S    14:05   0:02 [migration/4]
root          30  0.0  0.0      0     0 ?        S    14:05   0:00 [ksoftirqd/4]
root          32  0.0  0.0      0     0 ?        I<   14:05   0:00 [kworker/4:0H-events_highpri]
root          33  0.0  0.0      0     0 ?        S    14:05   0:00 [cpuhp/5]
root          34  0.0  0.0      0     0 ?        S    14:05   0:00 [idle_inject/5]
root          35  0.0  0.0      0     0 ?        S    14:05   0:01 [migration/5]
root          36  0.0  0.0      0     0 ?        S    14:05   0:00 [ksoftirqd/5]
root          38  0.0  0.0      0     0 ?        I<   14:05   0:00 [kworker/5:0H-events_highpri]
root          39  0.0  0.0      0     0 ?        S    14:05   0:00 [cpuhp/6]
root          40  0.0  0.0      0     0 ?        S    14:05   0:00 [idle_inject/6]
```

```
root          35  0.0  0.0      0     0 ?        S    14:05   0:01 [migration/5]
root          36  0.0  0.0      0     0 ?        S    14:05   0:00 [ksoftirqd/5]
root          38  0.0  0.0      0     0 ?        I<   14:05   0:00 [kworker/5:0H-events_highpri]
root          39  0.0  0.0      0     0 ?        S    14:05   0:00 [cpuhp/6]
root          40  0.0  0.0      0     0 ?        S    14:05   0:00 [idle_inject/6]
root          41  0.0  0.0      0     0 ?        S    14:05   0:01 [migration/6]
root          42  0.0  0.0      0     0 ?        S    14:05   0:00 [ksoftirqd/6]
root          44  0.0  0.0      0     0 ?        I<   14:05   0:00 [kworker/6:0H-events_highpri]
root          45  0.0  0.0      0     0 ?        S    14:05   0:00 [cpuhp/7]
root          46  0.0  0.0      0     0 ?        S    14:05   0:00 [idle_inject/7]
root          47  0.0  0.0      0     0 ?        S    14:05   0:00 [migration/7]
root          48  0.0  0.0      0     0 ?        S    14:05   0:00 [ksoftirqd/7]
root          50  0.0  0.0      0     0 ?        I<   14:05   0:00 [kworker/7:0H-events_highpri]
root          51  0.0  0.0      0     0 ?        S    14:05   0:00 [cpuhp/1]
root          52  0.0  0.0      0     0 ?        S    14:05   0:00 [idle_inject/1]
root          53  0.0  0.0      0     0 ?        S    14:05   0:00 [migration/1]
root          54  0.0  0.0      0     0 ?        S    14:05   0:00 [ksoftirqd/1]
root          56  0.0  0.0      0     0 ?        I<   14:05   0:00 [kworker/1:0H-events_highpri]
root          57  0.0  0.0      0     0 ?        S    14:05   0:00 [cpuhp/3]
root          58  0.0  0.0      0     0 ?        S    14:05   0:00 [idle_inject/3]
root          59  0.0  0.0      0     0 ?        S    14:05   0:00 [migration/3]
root          60  0.0  0.0      0     0 ?        S    14:05   0:00 [ksoftirqd/3]
root          62  0.0  0.0      0     0 ?        I<   14:05   0:00 [kworker/3:0H-events_highpri]
root          63  0.0  0.0      0     0 ?        S    14:05   0:00 [kdevtmpfs]
root          64  0.0  0.0      0     0 ?        I<   14:05   0:00 [kworker/R-inet_]
root          65  0.0  0.0      0     0 ?        S    14:05   0:00 [kauditd]
root          67  0.0  0.0      0     0 ?        S    14:05   0:00 [khungtaskd]
root          68  0.0  0.0      0     0 ?        S    14:05   0:00 [oom_reaper]
root          70  0.0  0.0      0     0 ?        I<   14:05   0:00 [kworker/R-write]
root          71  0.0  0.0      0     0 ?        S    14:05   0:01 [kcompactd0]
root          72  0.0  0.0      0     0 ?        SN   14:05   0:00 [ksmd]
root          73  0.0  0.0      0     0 ?        SN   14:05   0:00 [khugepaged]
root          74  0.0  0.0      0     0 ?        I<   14:05   0:00 [kworker/R-kinte]
root          75  0.0  0.0      0     0 ?        I<   14:05   0:00 [kworker/R-kbloc]
root          76  0.0  0.0      0     0 ?        I<   14:05   0:00 [kworker/R-blkcg]
root          78  0.0  0.0      0     0 ?        S    14:05   0:06 [irq/9-acpi]
```

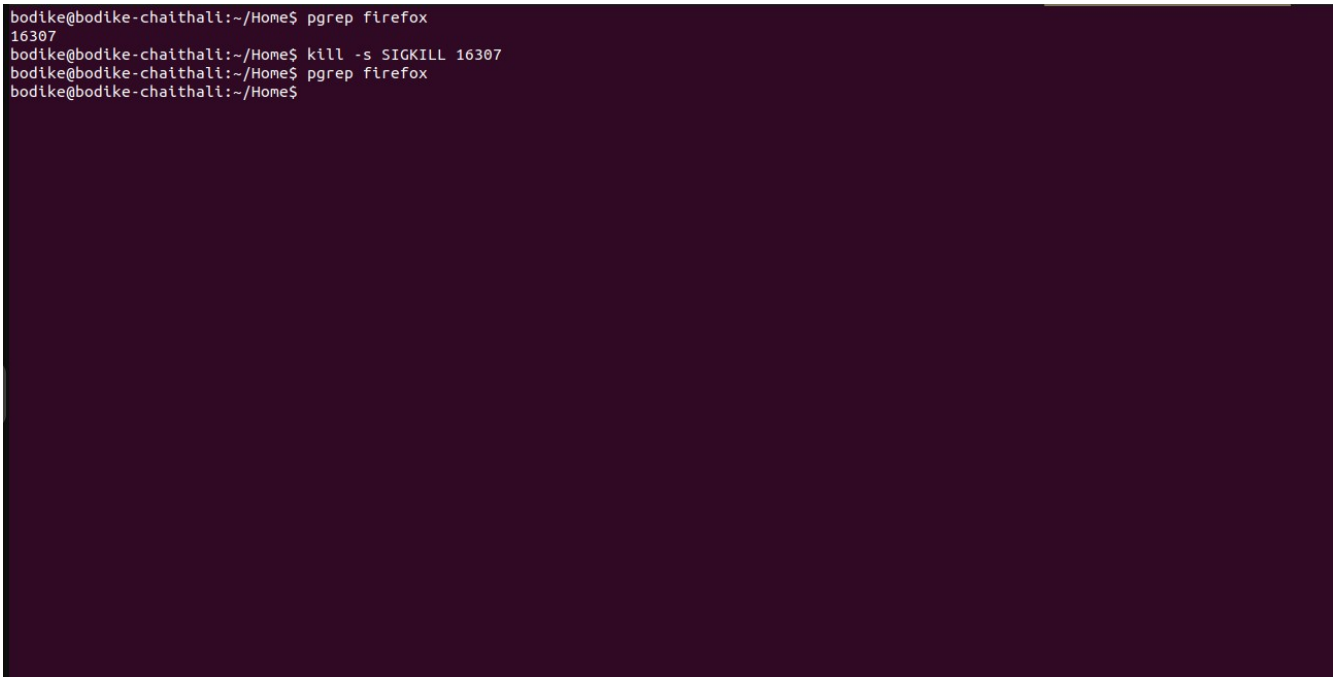
```
root          82  0.0  0.0      0     0 ?        I<   14:05   0:00 [kworker/R-tpm_d]
root          83  0.0  0.0      0     0 ?        I<   14:05   0:00 [kworker/R-ata_s]
root          85  0.0  0.0      0     0 ?        I<   14:05   0:00 [kworker/R-md]
root          86  0.0  0.0      0     0 ?        I<   14:05   0:00 [kworker/R-md_bi]
root          87  0.0  0.0      0     0 ?        I<   14:05   0:00 [kworker/R-edac-]
root          88  0.0  0.0      0     0 ?        I<   14:05   0:00 [kworker/R-devfr]
root          90  0.0  0.0      0     0 ?        S    14:05   0:00 [watchdogd]
root          91  0.0  0.0      0     0 ?        I<   14:05   0:00 [kworker/4:1H-events_highpri]
root          92  0.0  0.0      0     0 ?        S    14:05   0:00 [kswapd0]
root          93  0.0  0.0      0     0 ?        S    14:05   0:00 [ecryptfs-kthread]
root          94  0.0  0.0      0     0 ?        I<   14:05   0:00 [kworker/R-kthro]
root          96  0.0  0.0      0     0 ?        I<   14:05   0:00 [kworker/R-acpi_]
root          98  0.0  0.0      0     0 ?        S    14:05   0:00 [hwrng]
root          99  0.0  0.0      0     0 ?        I<   14:05   0:00 [kworker/R-hfi-u]
root         101  0.0  0.0      0     0 ?        I<   14:05   0:00 [kworker/R-mld]
root         102  0.0  0.0      0     0 ?        I<   14:05   0:00 [kworker/5:1H-events_highpri]
root         103  0.0  0.0      0     0 ?        I<   14:05   0:00 [kworker/R-ipv6_]
root         110  0.0  0.0      0     0 ?        I<   14:05   0:00 [kworker/R-kstrp]
root         127  0.0  0.0      0     0 ?        I<   14:05   0:00 [kworker/R-charg]
root         151  0.0  0.0      0     0 ?        I<   14:05   0:00 [kworker/0:1H-events_highpri]
root         152  0.0  0.0      0     0 ?        I<   14:05   0:00 [kworker/2:1H-events_highpri]
root         164  0.0  0.0      0     0 ?        I<   14:05   0:00 [kworker/6:1H]
root         166  0.0  0.0      0     0 ?        I<   14:05   0:00 [kworker/7:1H-events_highpri]
root         167  0.0  0.0      0     0 ?        I<   14:05   0:00 [kworker/1:1H-kblockd]
root         168  0.0  0.0      0     0 ?        I<   14:05   0:00 [kworker/3:1H-kblockd]
root         215  0.1  0.0      0     0 ?        S    14:05   0:23 [irq/160-VEN_04F3:00]
root         220  0.0  0.0      0     0 ?        I<   14:05   0:00 [kworker/R-USBC0]
root         222  0.0  0.0      0     0 ?        I<   14:05   0:00 [kworker/R-nvme-]
root         223  0.0  0.0      0     0 ?        I<   14:05   0:00 [kworker/R-nvme-]
root         224  0.0  0.0      0     0 ?        I<   14:05   0:00 [kworker/R-nvme-]
root         225  0.0  0.0      0     0 ?        I<   14:05   0:00 [kworker/R-nvme-]
root         237  0.0  0.0      0     0 ?        I<   14:05   0:00 [kworker/R-raid5]
root         269  0.0  0.0      0     0 ?        S    14:05   0:01 [jbd2/nvme0n1p3-8]
root         270  0.0  0.0      0     0 ?        I<   14:05   0:00 [kworker/R-ext4-]
root         316  0.0  1.3 148656 101632 ?       S<s  14:05   0:07 /lib/systemd/systemd-journald
root         376  0.0  0.0  27312  6656 ?        Ss   14:05   0:00 /lib/systemd/systemd-udevd
```

```
root        7821  0.0  0.0      0     0 ?        I    16:28   0:00 [kworker/2:2-events]
root        7846  0.0  0.0      0     0 ?        I    16:29   0:01 [kworker/1:2-events]
root        8272  0.0  0.0      0     0 ?        I    16:47   0:00 [kworker/7:1-i915-unordered]
root        8570  0.3  0.0      0     0 ?        R    16:59   0:15 [kworker/u16:0-events_unbound]
root        8625  0.0  0.0      0     0 ?        I    17:02   0:00 [kworker/4:0-events]
bodike      8829  0.0  0.0  10236  4224 pts/3    Ss+  17:11   0:00 bash
root        8881  0.0  0.0      0     0 ?        I    17:12   0:00 [kworker/0:1-events]
root        9353  0.0  0.0      0     0 ?        I    17:28   0:00 [kworker/5:2-events]
root        9752  0.0  0.0      0     0 ?        I    17:44   0:00 [kworker/6:1-cgroup_destroy]
root        9853  0.0  0.0      0     0 ?        I    17:49   0:00 [kworker/4:1-cgroup_destroy]
root        9856  0.0  0.0      0     0 ?        I<   17:50   0:00 [kworker/u17:2-rtw89_tx_wq]
root        9972  0.1  0.0      0     0 ?        I    18:02   0:01 [kworker/u16:2-flush-259:0]
root        9981  0.1  0.0      0     0 ?        I<   18:02   0:01 [kworker/u17:1-rtw89_tx_wq]
root       10021  0.0  0.0      0     0 ?        I    18:03   0:00 [kworker/7:0-i915-unordered]
root       10040  0.0  0.0      0     0 ?        I    18:05   0:00 [kworker/0:0-events]
bodike     10073  0.0  0.0  10348  4224 pts/2    Ss+  18:05   0:00 bash
root       10089  0.0  0.0      0     0 ?        I    18:06   0:00 [kworker/3:3-cgroup_destroy]
root       10098  0.0  0.0      0     0 ?        I    18:06   0:00 [kworker/u16:4-flush-259:0]
root       10104  0.0  0.0      0     0 ?        I    18:07   0:00 [kworker/1:3-cgroup_destroy]
root       10105  0.0  0.0      0     0 ?        I    18:07   0:00 [kworker/2:0-inet_frag_wq]
root       10246  0.1  0.0      0     0 ?        I    18:11   0:00 [kworker/u16:5-events_unbound]
root       10254  0.0  0.0      0     0 ?        I    18:12   0:00 [kworker/4:2-cgroup_destroy]
root       10289  0.0  0.0      0     0 ?        I<   18:12   0:00 [kworker/u17:0-i915_flip]
root       10344  0.0  0.0      0     0 ?        I    18:14   0:00 [kworker/1:0]
root       10345  0.0  0.0      0     0 ?        I    18:14   0:00 [kworker/1:1]
root       10346  0.0  0.0      0     0 ?        I    18:14   0:00 [kworker/2:1-cgroup_destroy]
root       10355  0.0  0.0      0     0 ?        I    18:14   0:00 [kworker/0:2-events]
bodike     10368  0.0  0.0  10236  4224 pts/4    Ss   18:15   0:00 bash
root       10383  0.0  0.0      0     0 ?        I    18:15   0:00 [kworker/5:1-cgroup_destroy]
root       10401  0.0  0.0      0     0 ?        I<   18:16   0:00 [kworker/u17:3-i915_flip]
root       10407  0.0  0.0      0     0 ?        I    18:16   0:00 [kworker/3:0]
root       10410  0.0  0.0      0     0 ?        I    18:16   0:00 [kworker/6:2]
bodike     10426  0.4  1.7 1461718432 139008 ?   Sl   18:17   0:00 /opt/google/chrome/chrome --type=renderer --crashpad-handler-pid=4681 --ena
bodike     10436  0.0  0.9 1459955052 71448 ?    Sl   18:17   0:00 /opt/google/chrome/chrome --type=renderer --crashpad-handler-pid=4681 --ena
bodike     10470  0.0  0.0  12672  3328 pts/4    R+   18:18   0:00 ps aux
bodike@bodike-chaithali:~/Home$
```

Q2)


bodike@bodike-chaithali:~/Home$ firefox
Gtk-Message: 18:32:02.843: Not loading module "atk-bridge": The functionality is provided by GTK natively. Please try to not load it.
bodike@bodike-chaithali:~/Home$ ps aux | grep firefox
bodike 15661 0.0 0.0 9080 2432 pts/5 R+ 18:33 0:00 grep firefox
bodike@bodike-chaithali:~/Home$

Q3)


bodike@bodike-chaithali:~/Home$ pgrep firefox
16307
bodike@bodike-chaithali:~/Home$ kill -s SIGKILL 16307
bodike@bodike-chaithali:~/Home$ pgrep firefox
bodike@bodike-chaithali:~/Home$

Q4)

```
bodike@bodike-chaithali:~/Home$ touch hello.c
bodike@bodike-chaithali:~/Home$ cat hello.c
bodike@bodike-chaithali:~/Home$ vim hello.c
bodike@bodike-chaithali:~/Home$ cat hello.c
#include<stdio.h>

int main () {
        printf("Hello World!\n");

        return 0;
}
bodike@bodike-chaithali:~/Home$ ls
hello.c
bodike@bodike-chaithali:~/Home$ gcc hello.c
bodike@bodike-chaithali:~/Home$ time ./hello
Hello World!

real    0m0.002s
user    0m0.000s
sys     0m0.002s
bodike@bodike-chaithali:~/Home$
```

Q5)

```
bodike@bodike-chaithali:~/Home$ ps -u $USER
    PID TTY          TIME CMD
   1456 ?        00:00:00 systemd
   1457 ?        00:00:00 (sd-pam)
   1464 ?        00:00:00 pipewire
   1465 ?        00:00:00 pipewire-media-
   1466 ?        00:00:02 pulseaudio
   1475 ?        00:00:00 gnome-keyring-d
   1496 ?        00:00:03 dbus-daemon
   1498 ?        00:00:00 xdg-document-po
   1501 ?        00:00:00 xdg-permission-
   1531 tty2     00:00:00 gdm-wayland-ses
   1534 tty2     00:00:00 gnome-session-b
   1585 ?        00:00:00 gnome-session-c
   1595 ?        00:00:00 gvfsd
   1603 ?        00:00:00 gvfsd-fuse
   1605 ?        00:00:00 gnome-session-b
   1626 ?        00:00:00 at-spi-bus-laun
   1632 ?        00:00:00 dbus-daemon
   1635 ?        00:08:15 gnome-shell
   1693 ?        00:00:00 gnome-shell-cal
   1699 ?        00:00:00 evolution-sourc
   1705 ?        00:00:00 goa-daemon
   1706 ?        00:00:00 gvfs-udisks2-vo
   1709 ?        00:00:00 evolution-calen
   1716 ?        00:00:00 gvfs-goa-volume
   1723 ?        00:00:00 goa-identity-se
   1729 ?        00:00:01 gvfs-afc-volume
   1738 ?        00:00:00 gvfs-gphoto2-vo
   1743 ?        00:00:00 dconf-service
   1751 ?        00:00:00 evolution-addre
   1752 ?        00:00:00 gvfs-mtp-volume
   1768 ?        00:00:30 nautilus
   1769 ?        00:00:07 gvfsd-trash
   1782 ?        00:00:00 gjs
   1784 ?        00:00:00 at-spi2-registr
   1799 ?        00:00:00 sh
   1800 ?        00:00:00 gsd-a11y-settin
```

Q6)

```
bodike@bodike-chaithali:~$ cal -j 2050
                              2050
          January                         February
 Su  Mo  Tu  We  Th  Fr  Sa    Su  Mo  Tu  We  Th  Fr  Sa
                      1                 32  33  34  35  36
  2   3   4   5   6   7   8    37  38  39  40  41  42  43
  9  10  11  12  13  14  15    44  45  46  47  48  49  50
 16  17  18  19  20  21  22    51  52  53  54  55  56  57
 23  24  25  26  27  28  29    58  59
 30  31

           March                           April
 Su  Mo  Tu  We  Th  Fr  Sa    Su  Mo  Tu  We  Th  Fr  Sa
         60  61  62  63  64                        91  92
 65  66  67  68  69  70  71    93  94  95  96  97  98  99
 72  73  74  75  76  77  78   100 101 102 103 104 105 106
 79  80  81  82  83  84  85   107 108 109 110 111 112 113
 86  87  88  89  90           114 115 116 117 118 119 120


            May                             June
 Su  Mo  Tu  We  Th  Fr  Sa    Su  Mo  Tu  We  Th  Fr  Sa
121 122 123 124 125 126 127                152 153 154 155
128 129 130 131 132 133 134   156 157 158 159 160 161 162
135 136 137 138 139 140 141   163 164 165 166 167 168 169
142 143 144 145 146 147 148   170 171 172 173 174 175 176
149 150 151                   177 178 179 180 181


           July                            August
 Su  Mo  Tu  We  Th  Fr  Sa    Su  Mo  Tu  We  Th  Fr  Sa
                     182 183       213 214 215 216 217 218
184 185 186 187 188 189 190   219 220 221 222 223 224 225
191 192 193 194 195 196 197   226 227 228 229 230 231 232
198 199 200 201 202 203 204   233 234 235 236 237 238 239
205 206 207 208 209 210 211   240 241 242 243
212
```

```
            May                             June
 Su  Mo  Tu  We  Th  Fr  Sa    Su  Mo  Tu  We  Th  Fr  Sa
121 122 123 124 125 126 127                152 153 154 155
128 129 130 131 132 133 134   156 157 158 159 160 161 162
135 136 137 138 139 140 141   163 164 165 166 167 168 169
142 143 144 145 146 147 148   170 171 172 173 174 175 176
149 150 151                   177 178 179 180 181

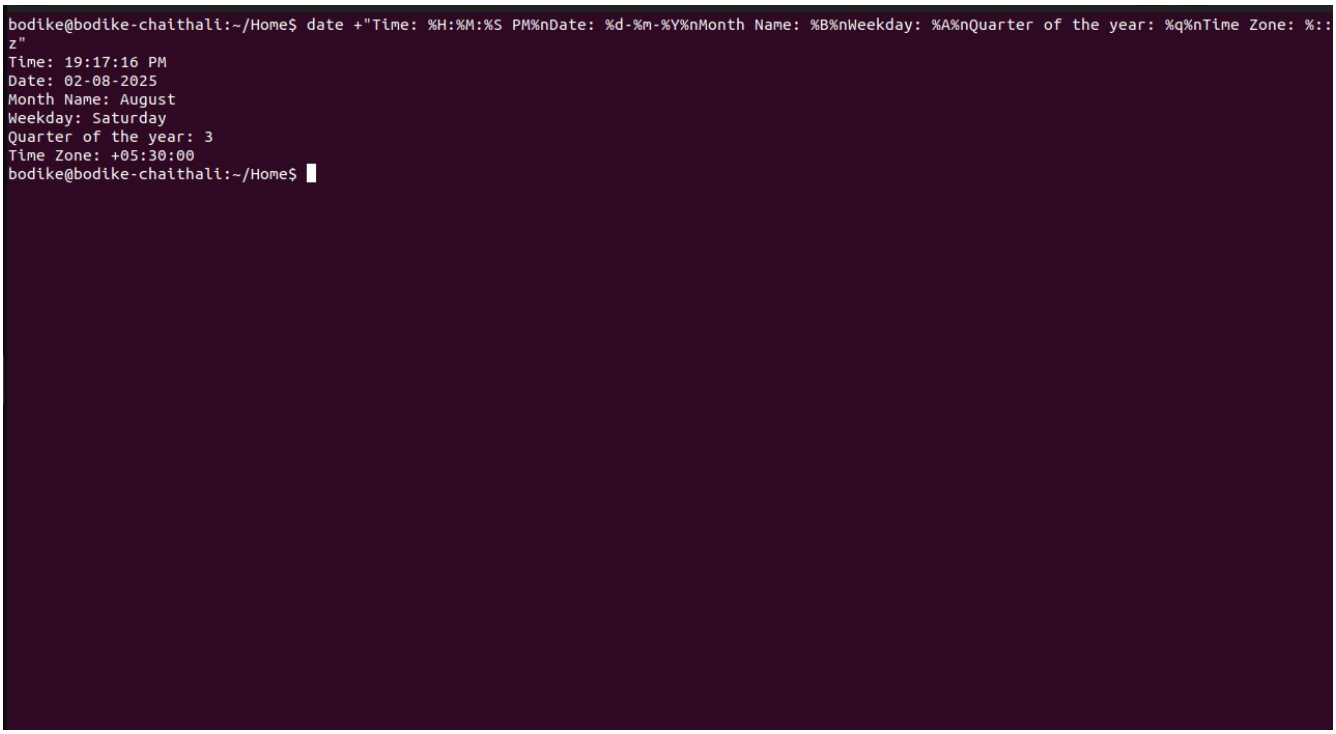

           July                            August
 Su  Mo  Tu  We  Th  Fr  Sa    Su  Mo  Tu  We  Th  Fr  Sa
                     182 183       213 214 215 216 217 218
184 185 186 187 188 189 190   219 220 221 222 223 224 225
191 192 193 194 195 196 197   226 227 228 229 230 231 232
198 199 200 201 202 203 204   233 234 235 236 237 238 239
205 206 207 208 209 210 211   240 241 242 243
212

         September                       October
 Su  Mo  Tu  We  Th  Fr  Sa    Su  Mo  Tu  We  Th  Fr  Sa
                 244 245 246                       274
247 248 249 250 251 252 253   275 276 277 278 279 280 281
254 255 256 257 258 259 260   282 283 284 285 286 287 288
261 262 263 264 265 266 267   289 290 291 292 293 294 295
268 269 270 271 272 273       296 297 298 299 300 301 302
                              303 304

         November                        December
 Su  Mo  Tu  We  Th  Fr  Sa    Su  Mo  Tu  We  Th  Fr  Sa
        305 306 307 308 309                   335 336 337
310 311 312 313 314 315 316   338 339 340 341 342 343 344
317 318 319 320 321 322 323   345 346 347 348 349 350 351
324 325 326 327 328 329 330   352 353 354 355 356 357 358
331 332 333 334               359 360 361 362 363 364 365

bodike@bodike-chaithali:~$
```

Q7)


bodike@bodike-chaithali:~/Home$ date +"Time: %H:%M:%S PM%nDate: %d-%m-%Y%nMonth Name: %B%nWeekday: %A%nQuarter of the year: %q%nTime Zone: %::
z"
Time: 19:17:16 PM
Date: 02-08-2025
Month Name: August
Weekday: Saturday
Quarter of the year: 3
Time Zone: +05:30:00
bodike@bodike-chaithali:~/Home$

Q8)

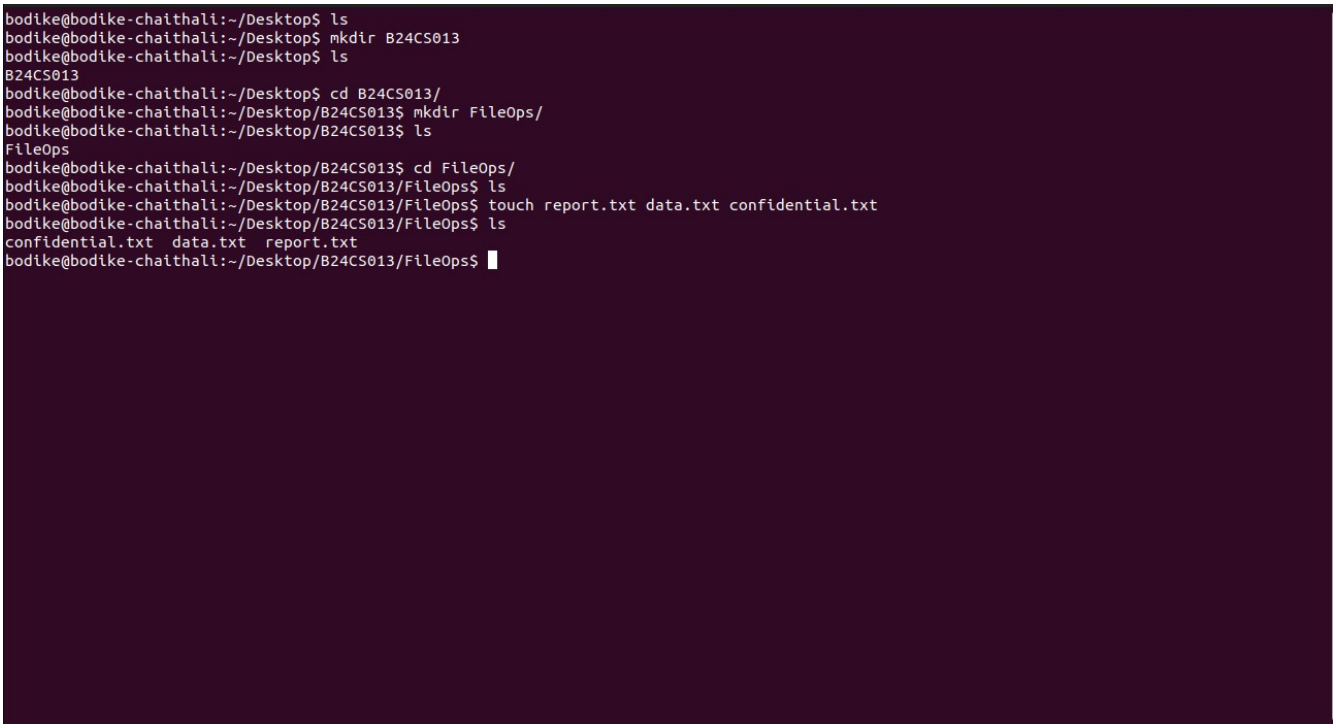
```
bodike@bodike-chaithali:~/Home$ date +"Time: %H:%M:%S PM%nDate: %d-%m-%Y%nMonth Name: %B%nWeekday: %A%nQuarter of the year: %q%nTime Zone: %::
z"
Time: 19:17:16 PM
Date: 02-08-2025
Month Name: August
Weekday: Saturday
Quarter of the year: 3
Time Zone: +05:30:00
bodike@bodike-chaithali:~/Home$ date -d "2 years 5 months 11 days 3 hours 22 minutes 20 seconds ago"
Thursday 13 January 2028 10:41:28 PM IST
bodike@bodike-chaithali:~/Home$ date -u -d "2 years 5 months 11 days 3 hours 22 minutes 20 seconds ago"
Thursday 13 January 2028 05:12:31 PM UTC
bodike@bodike-chaithali:~/Home$
```

Q9)

```
bodike@bodike-chaithali:~/Home$ date --date="2027-03-12 14:50:26" +"%H:%M:%S%nDate: %d/%m/%Y%nMonth Name: %B%nWeekday: %A (Weekday Number: %w)
%nQuarter of the year: %q%nTime Zone: %::z"
14:50:26
Date: 12/03/2027
Month Name: March
Weekday: Friday (Weekday Number: 5)
Quarter of the year: 1
Time Zone: +05:30:00
bodike@bodike-chaithali:~/Home$
```

Task 3:

Q1) and Q2)



Q3)

```
bodike@bodike-chaithali:~/Desktop/B24CS013/FileOps$ echo "This is report.txt file." > report.txt
bodike@bodike-chaithali:~/Desktop/B24CS013/FileOps$ cat report.txt
This is report.txt file.
bodike@bodike-chaithali:~/Desktop/B24CS013/FileOps$ cat data.txt
bodike@bodike-chaithali:~/Desktop/B24CS013/FileOps$ echo "This is data.txt file." > data.txt
bodike@bodike-chaithali:~/Desktop/B24CS013/FileOps$ cat data.txt
This is data.txt file.
bodike@bodike-chaithali:~/Desktop/B24CS013/FileOps$ cat confidential.txt
bodike@bodike-chaithali:~/Desktop/B24CS013/FileOps$ echo "This is confidential.txt file." > confidential.txt
bodike@bodike-chaithali:~/Desktop/B24CS013/FileOps$ cat confidential.txt
This is confidential.txt file.
bodike@bodike-chaithali:~/Desktop/B24CS013/FileOps$
```

Q4)

```
bodike@bodike-chaithali:~/Desktop/B24CS013/FileOps$ cat data.txt
This is data.txt file.
bodike@bodike-chaithali:~/Desktop/B24CS013/FileOps$ cat confidential.txt
bodike@bodike-chaithali:~/Desktop/B24CS013/FileOps$ echo "This is confidential.txt file." > confidential.txt
bodike@bodike-chaithali:~/Desktop/B24CS013/FileOps$ cat confidential.txt
This is confidential.txt file.
bodike@bodike-chaithali:~/Desktop/B24CS013/FileOps$ cp report.txt report_backup.txt
bodike@bodike-chaithali:~/Desktop/B24CS013/FileOps$ ls
confidential.txt  data.txt  report_backup.txt  report.txt
bodike@bodike-chaithali:~/Desktop/B24CS013/FileOps$ cat report_backup.txt
This is report.txt file.
bodike@bodike-chaithali:~/Desktop/B24CS013/FileOps$ mv data.txt dataset.csv
bodike@bodike-chaithali:~/Desktop/B24CS013/FileOps$ ls
confidential.txt  dataset.csv  report_backup.txt  report.txt
bodike@bodike-chaithali:~/Desktop/B24CS013/FileOps$ cat dataset.csv
This is data.txt file.
bodike@bodike-chaithali:~/Desktop/B24CS013/FileOps$ mkdir Secure
bodike@bodike-chaithali:~/Desktop/B24CS013/FileOps$ ls
confidential.txt  dataset.csv  report_backup.txt  report.txt  Secure
bodike@bodike-chaithali:~/Desktop/B24CS013/FileOps$ mv confidential.txt Secure/
bodike@bodike-chaithali:~/Desktop/B24CS013/FileOps$ ls
dataset.csv  report_backup.txt  report.txt  Secure
bodike@bodike-chaithali:~/Desktop/B24CS013/FileOps$ ls Secure
confidential.txt
bodike@bodike-chaithali:~/Desktop/B24CS013/FileOps$ cd Secure/
bodike@bodike-chaithali:~/Desktop/B24CS013/FileOps/Secure$ cat confidential.txt
This is confidential.txt file.
bodike@bodike-chaithali:~/Desktop/B24CS013/FileOps/Secure$ cd ..
bodike@bodike-chaithali:~/Desktop/B24CS013/FileOps$ cd ..
bodike@bodike-chaithali:~/Desktop/B24CS013$ ls
FileOps
bodike@bodike-chaithali:~/Desktop/B24CS013$ mv FileOps SystemFiles
bodike@bodike-chaithali:~/Desktop/B24CS013$ ls
SystemFiles
bodike@bodike-chaithali:~/Desktop/B24CS013$ cd SystemFiles/
bodike@bodike-chaithali:~/Desktop/B24CS013/SystemFiles$ ls
dataset.csv  report_backup.txt  report.txt  Secure
bodike@bodike-chaithali:~/Desktop/B24CS013/SystemFiles$
```

Q5)

```
bodike@bodike-chaithali:~/Desktop/B24CS013/SystemFiles$ chmod 600 report.txt
bodike@bodike-chaithali:~/Desktop/B24CS013/SystemFiles$ chmod 444 report_backup.txt
bodike@bodike-chaithali:~/Desktop/B24CS013/SystemFiles$ chmod 111 dataset.csv
bodike@bodike-chaithali:~/Desktop/B24CS013/SystemFiles$ ls -l
total 16
---x--x--x 1 bodike bodike   23 Aug  2 20:00 dataset.csv
-r--r--r-- 1 bodike bodike   25 Aug  2 20:02 report_backup.txt
-rw------- 1 bodike bodike   25 Aug  2 20:00 report.txt
drwxrwxr-x 2 bodike bodike 4096 Aug  2 20:07 Secure
bodike@bodike-chaithali:~/Desktop/B24CS013/SystemFiles$ cd Secure/
bodike@bodike-chaithali:~/Desktop/B24CS013/SystemFiles/Secure$ chmod 600 confidential.txt
bodike@bodike-chaithali:~/Desktop/B24CS013/SystemFiles/Secure$ ls -l
total 4
-rw------- 1 bodike bodike 31 Aug  2 20:01 confidential.txt
bodike@bodike-chaithali:~/Desktop/B24CS013/SystemFiles/Secure$ cd ..
bodike@bodike-chaithali:~/Desktop/B24CS013/SystemFiles$
```

Q6)

```
bodike@bodike-chaithali:~/Desktop/B24CS013/SystemFiles$ chmod 600 report.txt
bodike@bodike-chaithali:~/Desktop/B24CS013/SystemFiles$ chmod 444 report_backup.txt
bodike@bodike-chaithali:~/Desktop/B24CS013/SystemFiles$ chmod 111 dataset.csv
bodike@bodike-chaithali:~/Desktop/B24CS013/SystemFiles$ ls -l
total 16
---x--x--x 1 bodike bodike   23 Aug  2 20:00 dataset.csv
-r--r--r-- 1 bodike bodike   25 Aug  2 20:02 report_backup.txt
-rw------- 1 bodike bodike   25 Aug  2 20:00 report.txt
drwxrwxr-x 2 bodike bodike 4096 Aug  2 20:07 Secure
bodike@bodike-chaithali:~/Desktop/B24CS013/SystemFiles$ cd Secure/
bodike@bodike-chaithali:~/Desktop/B24CS013/SystemFiles/Secure$ chmod 600 confidential.txt
bodike@bodike-chaithali:~/Desktop/B24CS013/SystemFiles/Secure$ ls -l
total 4
-rw------- 1 bodike bodike 31 Aug  2 20:01 confidential.txt
bodike@bodike-chaithali:~/Desktop/B24CS013/SystemFiles/Secure$ cd ..
bodike@bodike-chaithali:~/Desktop/B24CS013/SystemFiles$ ls -l report.txt report_backup.txt dataset.csv Secure/confidential.txt > permissions.t
xt
bodike@bodike-chaithali:~/Desktop/B24CS013/SystemFiles$ cat permissions.txt
---x--x--x 1 bodike bodike 23 Aug  2 20:00 dataset.csv
-r--r--r-- 1 bodike bodike 25 Aug  2 20:02 report_backup.txt
-rw------- 1 bodike bodike 25 Aug  2 20:00 report.txt
-rw------- 1 bodike bodike 31 Aug  2 20:01 Secure/confidential.txt
bodike@bodike-chaithali:~/Desktop/B24CS013/SystemFiles$ ls -l report.txt report_backup.txt dataset.csv Secure/confidential.txt
---x--x--x 1 bodike bodike 23 Aug  2 20:00 dataset.csv
-r--r--r-- 1 bodike bodike 25 Aug  2 20:02 report_backup.txt
-rw------- 1 bodike bodike 25 Aug  2 20:00 report.txt
-rw------- 1 bodike bodike 31 Aug  2 20:01 Secure/confidential.txt
bodike@bodike-chaithali:~/Desktop/B24CS013/SystemFiles$
```

Task 4:

Q1)

```
bodike@bodike-chaithali:~$ ps -u $USER
    PID TTY          TIME CMD
   1456 ?        00:00:00 systemd
   1457 ?        00:00:00 (sd-pam)
   1464 ?        00:00:00 pipewire
   1465 ?        00:00:00 pipewire-media-
   1466 ?        00:00:03 pulseaudio
   1475 ?        00:00:00 gnome-keyring-d
   1496 ?        00:00:04 dbus-daemon
   1498 ?        00:00:00 xdg-document-po
   1501 ?        00:00:00 xdg-permission-
   1531 tty2     00:00:00 gdm-wayland-ses
   1534 tty2     00:00:00 gnome-session-b
   1585 ?        00:00:00 gnome-session-c
   1595 ?        00:00:00 gvfsd
   1603 ?        00:00:00 gvfsd-fuse
   1605 ?        00:00:00 gnome-session-b
   1626 ?        00:00:00 at-spi-bus-laun
   1632 ?        00:00:00 dbus-daemon
   1635 ?        00:12:29 gnome-shell
   1693 ?        00:00:00 gnome-shell-cal
   1699 ?        00:00:00 evolution-sourc
   1705 ?        00:00:00 goa-daemon
   1706 ?        00:00:00 gvfs-udisks2-vo
   1709 ?        00:00:00 evolution-calen
   1716 ?        00:00:00 gvfs-goa-volume
   1723 ?        00:00:00 goa-identity-se
   1729 ?        00:00:01 gvfs-afc-volume
   1738 ?        00:00:00 gvfs-gphoto2-vo
   1743 ?        00:00:00 dconf-service
   1751 ?        00:00:00 evolution-addre
   1752 ?        00:00:00 gvfs-mtp-volume
   1769 ?        00:00:07 gvfsd-trash
   1782 ?        00:00:00 gjs
   1784 ?        00:00:00 at-spi2-registr
   1799 ?        00:00:00 sh
```

```
   1886 ?        00:00:00 evolution-alarm
   1887 ?        00:00:00 ibus-portal
   1949 ?        00:00:03 snap-store
   1980 ?        00:00:20 ibus-engine-sim
   2044 ?        00:00:00 gsd-printer
   2058 ?        00:00:01 xdg-desktop-por
   2062 ?        00:00:07 tracker-miner-f
   2069 ?        00:00:03 xdg-desktop-por
   2099 ?        00:00:00 gjs
   2136 ?        00:00:00 xdg-desktop-por
   2220 ?        00:00:00 gvfsd-metadata
   2270 ?        00:00:00 snapd-desktop-i
   2332 ?        00:00:00 snapd-desktop-i
   4580 ?        00:00:00 update-notifier
   4618 ?        00:01:29 gnome-terminal-
   4731 ?        00:00:01 Xwayland
   4774 ?        00:00:00 gsd-xsettings
   4823 ?        00:00:00 ibus-x11
   5120 ?        00:00:00 evinced
   6616 ?        00:00:41 evince
   6856 ?        00:00:11 gjs
   7783 ?        00:00:00 gvfsd-network
   7799 ?        00:00:00 gvfsd-dnssd
  10896 ?        00:00:00 snap
  11267 pts/5    00:00:00 bash
  18089 ?        00:00:07 nautilus
  18205 ?        00:00:00 oosplash
  18221 ?        00:00:16 soffice.bin
  18709 ?        00:00:00 gnome-terminal
  18712 ?        00:00:00 gnome-terminal.
  18717 pts/0    00:00:00 bash
  20461 ?        00:00:03 evince
  21780 pts/1    00:00:00 bash
  21978 pts/2    00:00:00 bash
  22004 pts/2    00:00:00 ps
bodike@bodike-chaithali:~$
```

Q2)

```
bodike@bodike-chaithali:~$ ps aux --sort=-%mem | head -n 6

USER         PID %CPU %MEM    VSZ   RSS TTY      STAT START   TIME COMMAND
bodike     18221  0.3  6.0 1511296 475196 ?      Sl   18:57   0:18 /usr/lib/libreoffice/program/soffice.bin --writer file:///home/bodike/Doc
ents/Ass.odt
bodike     22035  4.0  4.2 34388420 333208 ?     SLl  20:26   0:05 /opt/google/chrome/chrome
bodike      1635  3.3  4.2 5880272 330276 ?      Ssl  14:05  12:41 /usr/bin/gnome-shell
bodike     22268 11.5  3.2 1461756036 257248 ?   Sl   20:27   0:15 /opt/google/chrome/chrome --type=renderer --crashpad-handler-pid=22043 --
able-crash-reporter=afe445fd-fb76-448d-ab7c-6f9086a80e3f, --change-stack-guard-on-fork=enable --ozone-platform=wayland --lang=en-GB --num-ra
er-threads=4 --enable-main-frame-before-activation --renderer-client-id=13 --time-ticks-at-unix-epoch=-1754124377230151 --launch-time-ticks=
245136654 --shared-files=v8_context_snapshot_data:100 --metrics-shmem-handle=4,i,8705618808265611410,15139744613126290436,2097152 --field-tr
l-handle=3,i,2551573121741561986,2211497720720656676,262144 --disable-features=EyeDropper --variations-seed-version=20250801-130007.450000
bodike     22091  2.7  2.2 34666240 174020 ?     Sl   20:26   0:03 /opt/google/chrome/chrome --type=gpu-process --ozone-platform=wayland --c
shpad-handler-pid=22043 --enable-crash-reporter=afe445fd-fb76-448d-ab7c-6f9086a80e3f, --change-stack-guard-on-fork=enable --gpu-preferences=
AAAAAAAAAgAAAEAAAAAAAAAAAAAAAAAABgAAEAAAAAAAAAAAAAAAAAAAAAAAAAAAAAAAAAAAAAAAAAAAAAAAABAAAAAAAAAAEAAAAAAAAAIAAAAAAAAAAgAAAAAAAA --shared-file
--metrics-shmem-handle=4,i,17855435638386186065,15047600428469152481,262144 --field-trial-handle=3,i,2551573121741561986,2211497720720656676
62144 --disable-features=EyeDropper --variations-seed-version=20250801-130007.450000
bodike@bodike-chaithali:~$
```

Q3)

```
  10896 ?        00:00:00 snap
  11267 pts/5    00:00:00 bash
  18089 ?        00:00:07 nautilus
  18205 ?        00:00:00 oosplash
  18221 ?        00:00:16 soffice.bin
  18709 ?        00:00:00 gnome-terminal
  18712 ?        00:00:00 gnome-terminal.
  18717 pts/0    00:00:00 bash
  20461 ?        00:00:03 evince
  21780 pts/1    00:00:00 bash
  21978 pts/2    00:00:00 bash
  22004 pts/2    00:00:00 ps
bodike@bodike-chaithali:~$ ps aux --sort=-%mem | head -n 6

USER         PID %CPU %MEM    VSZ   RSS TTY      STAT START   TIME COMMAND
bodike     18221  0.3  6.0 1511296 475196 ?      Sl   18:57   0:18 /usr/lib/libreoffice/program/soffice.bin --writer file:///home/bodike/Docum
ents/Ass.odt
bodike     22035  4.0  4.2 34388420 333208 ?     SLl  20:26   0:05 /opt/google/chrome/chrome
bodike      1635  3.3  4.2 5880272 330276 ?      Ssl  14:05  12:41 /usr/bin/gnome-shell
bodike     22268 11.5  3.2 1461756036 257248 ?   Sl   20:27   0:15 /opt/google/chrome/chrome --type=renderer --crashpad-handler-pid=22043 --en
able-crash-reporter=afe445fd-fb76-448d-ab7c-6f9086a80e3f, --change-stack-guard-on-fork=enable --ozone-platform=wayland --lang=en-GB --num-rast
er-threads=4 --enable-main-frame-before-activation --renderer-client-id=13 --time-ticks-at-unix-epoch=-1754124377230151 --launch-time-ticks=22
245136654 --shared-files=v8_context_snapshot_data:100 --metrics-shmem-handle=4,i,8705618808265611410,15139744613126290436,2097152 --field-tria
l-handle=3,i,2551573121741561986,2211497720720656676,262144 --disable-features=EyeDropper --variations-seed-version=20250801-130007.450000
bodike     22091  2.7  2.2 34666240 174020 ?     Sl   20:26   0:03 /opt/google/chrome/chrome --type=gpu-process --ozone-platform=wayland --cra
shpad-handler-pid=22043 --enable-crash-reporter=afe445fd-fb76-448d-ab7c-6f9086a80e3f, --change-stack-guard-on-fork=enable --gpu-preferences=UA
AAAAAAAAAgAAAEAAAAAAAAAAAAAAAAAABgAAEAAAAAAAAAAAAAAAAAAAAAAAAAAAAAAAAAAAAAAAAAAAAAAAABAAAAAAAAAAEAAAAAAAAAIAAAAAAAAAAgAAAAAAAA --shared-files
--metrics-shmem-handle=4,i,17855435638386186065,15047600428469152481,262144 --field-trial-handle=3,i,2551573121741561986,2211497720720656676,2
62144 --disable-features=EyeDropper --variations-seed-version=20250801-130007.450000
bodike@bodike-chaithali:~$ hostname
bodike-chaithali
bodike@bodike-chaithali:~$ uname -o
GNU/Linux
```

Q4)

```
bodike@bodike-chaithali:~$ date
Saturday 02 August 2025 09:01:02 PM IST
bodike@bodike-chaithali:~$
```

Q5)

```
bodike@bodike-chaithali:~$ env > env.txt
bodike@bodike-chaithali:~$ cat env.txt
SHELL=/bin/bash
SESSION_MANAGER=local/bodike-chaithali:@/tmp/.ICE-unix/1605,unix/bodike-chaithali:/tmp/.ICE-unix/1605
QT_ACCESSIBILITY=1
COLORTERM=truecolor
XDG_CONFIG_DIRS=/etc/xdg/xdg-ubuntu:/etc/xdg
SSH_AGENT_LAUNCHER=gnome-keyring
XDG_MENU_PREFIX=gnome-
GNOME_DESKTOP_SESSION_ID=this-is-deprecated
LANGUAGE=en_IN:en
GNOME_SHELL_SESSION_MODE=ubuntu
SSH_AUTH_SOCK=/run/user/1001/keyring/ssh
XMODIFIERS=@im=ibus
DESKTOP_SESSION=ubuntu
GTK_MODULES=gail:atk-bridge
PWD=/home/bodike
LOGNAME=bodike
XDG_SESSION_DESKTOP=ubuntu
XDG_SESSION_TYPE=wayland
SYSTEMD_EXEC_PID=4618
XAUTHORITY=/run/user/1001/.mutter-Xwaylandauth.WD0HA3
HOME=/home/bodike
USERNAME=bodike
IM_CONFIG_PHASE=1
LANG=en_IN
XDG_CURRENT_DESKTOP=ubuntu:GNOME
VTE_VERSION=6800
WAYLAND_DISPLAY=wayland-0
GNOME_TERMINAL_SCREEN=/org/gnome/Terminal/screen/1081c165_0079_45c5_bcbd_8aeb26a37bb5
GNOME_SETUP_DISPLAY=:1
XDG_SESSION_CLASS=user
TERM=xterm-256color
USER=bodike
GNOME_TERMINAL_SERVICE=:1.112
DISPLAY=:0
```

Q6)

```
bodike@bodike-chaithali:~$ ps -u $USER
    PID TTY          TIME CMD
   1456 ?        00:00:00 systemd
   1457 ?        00:00:00 (sd-pam)
   1464 ?        00:00:00 pipewire
   1465 ?        00:00:00 pipewire-media-
   1466 ?        00:00:03 pulseaudio
   1475 ?        00:00:00 gnome-keyring-d
   1496 ?        00:00:04 dbus-daemon
   1498 ?        00:00:00 xdg-document-po
   1501 ?        00:00:00 xdg-permission-
   1531 tty2     00:00:00 gdm-wayland-ses
   1534 tty2     00:00:00 gnome-session-b
   1585 ?        00:00:00 gnome-session-c
   1595 ?        00:00:00 gvfsd
   1603 ?        00:00:00 gvfsd-fuse
   1605 ?        00:00:00 gnome-session-b
   1626 ?        00:00:00 at-spi-bus-laun
   1632 ?        00:00:00 dbus-daemon
   1635 ?        00:12:29 gnome-shell
   1693 ?        00:00:00 gnome-shell-cal
   1699 ?        00:00:00 evolution-sourc
   1705 ?        00:00:00 goa-daemon
   1706 ?        00:00:00 gvfs-udisks2-vo
   1709 ?        00:00:00 evolution-calen
   1716 ?        00:00:00 gvfs-goa-volume
   1723 ?        00:00:00 goa-identity-se
   1729 ?        00:00:01 gvfs-afc-volume
   1738 ?        00:00:00 gvfs-gphoto2-vo
   1743 ?        00:00:00 dconf-service
   1751 ?        00:00:00 evolution-addre
   1752 ?        00:00:00 gvfs-mtp-volume
   1769 ?        00:00:07 gvfsd-trash
   1782 ?        00:00:00 gjs
   1784 ?        00:00:00 at-spi2-registr
   1799 ?        00:00:00 sh
```

```
   1886 ?        00:00:00 evolution-alarm
   1887 ?        00:00:00 ibus-portal
   1949 ?        00:00:03 snap-store
   1980 ?        00:00:20 ibus-engine-sim
   2044 ?        00:00:00 gsd-printer
   2058 ?        00:00:01 xdg-desktop-por
   2062 ?        00:00:07 tracker-miner-f
   2069 ?        00:00:03 xdg-desktop-por
   2099 ?        00:00:00 gjs
   2136 ?        00:00:00 xdg-desktop-por
   2220 ?        00:00:00 gvfsd-metadata
   2270 ?        00:00:00 snapd-desktop-i
   2332 ?        00:00:00 snapd-desktop-i
   4580 ?        00:00:00 update-notifier
   4618 ?        00:01:29 gnome-terminal-
   4731 ?        00:00:01 Xwayland
   4774 ?        00:00:00 gsd-xsettings
   4823 ?        00:00:00 ibus-x11
   5120 ?        00:00:00 evinced
   6616 ?        00:00:41 evince
   6856 ?        00:00:11 gjs
   7783 ?        00:00:00 gvfsd-network
   7799 ?        00:00:00 gvfsd-dnssd
  10896 ?        00:00:00 snap
  11267 pts/5    00:00:00 bash
  18089 ?        00:00:07 nautilus
  18205 ?        00:00:00 oosplash
  18221 ?        00:00:16 soffice.bin
  18709 ?        00:00:00 gnome-terminal
  18712 ?        00:00:00 gnome-terminal.
  18717 pts/0    00:00:00 bash
  20461 ?        00:00:03 evince
  21780 pts/1    00:00:00 bash
  21978 pts/2    00:00:00 bash
  22004 pts/2    00:00:00 ps
bodike@bodike-chaithali:~$
```

Q7)

```
bodike@bodike-chaithali:~$ ps aux --sort=-%mem | head -n 6

USER         PID %CPU %MEM    VSZ   RSS TTY      STAT START   TIME COMMAND
bodike     18221  0.3  6.0 1511296 475196 ?      Sl   18:57   0:18 /usr/lib/libreoffice/program/soffice.bin --writer file:///home/bodike/Doc
ents/Ass.odt
bodike     22035  4.0  4.2 34388420 333208 ?     SLl  20:26   0:05 /opt/google/chrome/chrome
bodike      1635  3.3  4.2 5880272 330276 ?      Ssl  14:05  12:41 /usr/bin/gnome-shell
bodike     22268 11.5  3.2 1461756036 257248 ?   Sl   20:27   0:15 /opt/google/chrome/chrome --type=renderer --crashpad-handler-pid=22043 --
able-crash-reporter=afe445fd-fb76-448d-ab7c-6f9086a80e3f, --change-stack-guard-on-fork=enable --ozone-platform=wayland --lang=en-GB --num-ra
er-threads=4 --enable-main-frame-before-activation --renderer-client-id=13 --time-ticks-at-unix-epoch=-1754124377230151 --launch-time-ticks=
245136654 --shared-files=v8_context_snapshot_data:100 --metrics-shmem-handle=4,i,8705618808265611410,15139744613126290436,2097152 --field-tr
l-handle=3,i,2551573121741561986,221149720720656676,262144 --disable-features=EyeDropper --variations-seed-version=20250801-130007.450000
bodike     22091  2.7  2.2 34666240 174020 ?     Sl   20:26   0:03 /opt/google/chrome/chrome --type=gpu-process --ozone-platform=wayland --c
shpad-handler-pid=22043 --enable-crash-reporter=afe445fd-fb76-448d-ab7c-6f9086a80e3f, --change-stack-guard-on-fork=enable --gpu-preferences=
AAAAAAAAAgAAAEAAAAAAAAAAAAAAAAAABgAAEAAAAAAAAAAAAAAAAAAAAAAAAAAAAAAAAAAAAAAAAAAAAAAAAAAABAAAAAAAAAAEAAAAAAAAAAIAAAAAAAAAAgAAAAAAAAA --shared-file
--metrics-shmem-handle=4,i,17855435638386186065,15047600428469152481,262144 --field-trial-handle=3,i,2551573121741561986,221149772072065676
62144 --disable-features=EyeDropper --variations-seed-version=20250801-130007.450000
bodike@bodike-chaithali:~$
```

Q8)

```
  10896 ?        00:00:00 snap
  11267 pts/5    00:00:00 bash
  18089 ?        00:00:07 nautilus
  18205 ?        00:00:00 oosplash
  18221 ?        00:00:16 soffice.bin
  18709 ?        00:00:00 gnome-terminal
  18712 ?        00:00:00 gnome-terminal.
  18717 pts/0    00:00:00 bash
  20461 ?        00:00:03 evince
  21780 pts/1    00:00:00 bash
  21978 pts/2    00:00:00 bash
  22004 pts/2    00:00:00 ps
bodike@bodike-chaithali:~$ ps aux --sort=-%mem | head -n 6

USER         PID %CPU %MEM    VSZ   RSS TTY      STAT START   TIME COMMAND
bodike     18221  0.3  6.0 1511296 475196 ?      Sl   18:57   0:18 /usr/lib/libreoffice/program/soffice.bin --writer file:///home/bodike/Docum
ents/Ass.odt
bodike     22035  4.0  4.2 34388420 333208 ?     SLl  20:26   0:05 /opt/google/chrome/chrome
bodike      1635  3.3  4.2 5880272 330276 ?      Ssl  14:05  12:41 /usr/bin/gnome-shell
bodike     22268 11.5  3.2 1461756036 257248 ?   Sl   20:27   0:15 /opt/google/chrome/chrome --type=renderer --crashpad-handler-pid=22043 --en
able-crash-reporter=afe445fd-fb76-448d-ab7c-6f9086a80e3f, --change-stack-guard-on-fork=enable --ozone-platform=wayland --lang=en-GB --num-rast
er-threads=4 --enable-main-frame-before-activation --renderer-client-id=13 --time-ticks-at-unix-epoch=-1754124377230151 --launch-time-ticks=22
245136654 --shared-files=v8_context_snapshot_data:100 --metrics-shmem-handle=4,i,8705618808265611410,15139744613126290436,2097152 --field-tria
l-handle=3,i,2551573121741561986,221149772072065676,262144 --disable-features=EyeDropper --variations-seed-version=20250801-130007.450000
bodike     22091  2.7  2.2 34666240 174020 ?     Sl   20:26   0:03 /opt/google/chrome/chrome --type=gpu-process --ozone-platform=wayland --cra
shpad-handler-pid=22043 --enable-crash-reporter=afe445fd-fb76-448d-ab7c-6f9086a80e3f, --change-stack-guard-on-fork=enable --gpu-preferences=UA
AAAAAAAAgAAAEAAAAAAAAAAAAAAAAAABgAAEAAAAAAAAAAAAAAAAAAAAAAAAAAAAAAAAAAAAAAAAAAAAAAAAAAABAAAAAAAAAAEAAAAAAAAAAIAAAAAAAAAAgAAAAAAAAA --shared-files
--metrics-shmem-handle=4,i,17855435638386186065,15047600428469152481,262144 --field-trial-handle=3,i,2551573121741561986,221149772072065676,2
62144 --disable-features=EyeDropper --variations-seed-version=20250801-130007.450000
bodike@bodike-chaithali:~$ hostname
bodike-chaithali
bodike@bodike-chaithali:~$ uname -o
GNU/Linux
```

Q9)

```
bodike@bodike-chaithali:~$ date
Saturday 02 August 2025 09:01:02 PM IST
bodike@bodike-chaithali:~$
```

Q10)

```
bodike@bodike-chaithali:~$ env > env.txt
bodike@bodike-chaithali:~$ cat env.txt
SHELL=/bin/bash
SESSION_MANAGER=local/bodike-chaithali:@/tmp/.ICE-unix/1605,unix/bodike-chaithali:/tmp/.ICE-unix/1605
QT_ACCESSIBILITY=1
COLORTERM=truecolor
XDG_CONFIG_DIRS=/etc/xdg/xdg-ubuntu:/etc/xdg
SSH_AGENT_LAUNCHER=gnome-keyring
XDG_MENU_PREFIX=gnome-
GNOME_DESKTOP_SESSION_ID=this-is-deprecated
LANGUAGE=en_IN:en
GNOME_SHELL_SESSION_MODE=ubuntu
SSH_AUTH_SOCK=/run/user/1001/keyring/ssh
XMODIFIERS=@im=ibus
DESKTOP_SESSION=ubuntu
GTK_MODULES=gail:atk-bridge
PWD=/home/bodike
LOGNAME=bodike
XDG_SESSION_DESKTOP=ubuntu
XDG_SESSION_TYPE=wayland
SYSTEMD_EXEC_PID=4618
XAUTHORITY=/run/user/1001/.mutter-Xwaylandauth.WD0HA3
HOME=/home/bodike
USERNAME=bodike
IM_CONFIG_PHASE=1
LANG=en_IN
XDG_CURRENT_DESKTOP=ubuntu:GNOME
VTE_VERSION=6800
WAYLAND_DISPLAY=wayland-0
GNOME_TERMINAL_SCREEN=/org/gnome/Terminal/screen/1081c165_0079_45c5_bcbd_8aeb26a37bb5
GNOME_SETUP_DISPLAY=:1
XDG_SESSION_CLASS=user
TERM=xterm-256color
USER=bodike
GNOME_TERMINAL_SERVICE=:1.112
DISPLAY=:0
```

Task 5:

Q1)

```
bodike@bodike-chaithali:~$ echo "This is line which creates file with in the line." > greetings.txt
bodike@bodike-chaithali:~$ cat greetings.txt
This is line which creates file with in the line.
bodike@bodike-chaithali:~$
```

Q2)

```
bodike@bodike-chaithali:~$ echo "This is line which creates file with in the line." > greetings.txt
bodike@bodike-chaithali:~$ cat greetings.txt
This is line which creates file with in the line.
bodike@bodike-chaithali:~$ echo "Appened one line to the greetings file." >> greetings.txt
bodike@bodike-chaithali:~$ cat greetings.txt
This is line which creates file with in the line.
Appened one line to the greetings file.
bodike@bodike-chaithali:~$ echo "Appened another line to the greetings file." >> greetings.txt
bodike@bodike-chaithali:~$ cat greetings.txt
This is line which creates file with in the line.
Appened one line to the greetings file.
Appened another line to the greetings file.
bodike@bodike-chaithali:~$
```

Q3)

```
bodike@bodike-chaithali:~$ cat greetings.txt
This is line which creates file with in the line.
Appened one line to the greetings file.
Appened another line to the greetings file.
bodike@bodike-chaithali:~$
```

Q4)

```
bodike@bodike-chaithali:~$ for i in {4..23}; do echo "This is line $i" >> greetings.txt; done
bodike@bodike-chaithali:~$ cat greetings.txt
This is the line which creates file with in the line.
Appened one line to the greetings file.
Appened another line to the greetings file.
This is line 4
This is line 5
This is line 6
This is line 7
This is line 8
This is line 9
This is line 10
This is line 11
This is line 12
This is line 13
This is line 14
This is line 15
This is line 16
This is line 17
This is line 18
This is line 19
This is line 20
This is line 21
This is line 22
This is line 23
bodike@bodike-chaithali:~$
```

Q5)

```
bodike@bodike-chaithali:~$ for i in {4..23}; do echo "This is line $i" >> greetings.txt; done
bodike@bodike-chaithali:~$ cat greetings.txt
This is the line which creates file with in the line.
Appened one line to the greetings file.
Appened another line to the greetings file.
This is line 4
This is line 5
This is line 6
This is line 7
This is line 8
This is line 9
This is line 10
This is line 11
This is line 12
This is line 13
This is line 14
This is line 15
This is line 16
This is line 17
This is line 18
This is line 19
This is line 20
This is line 21
This is line 22
This is line 23
bodike@bodike-chaithali:~$ head -n 5 greetings.txt
This is the line which creates file with in the line.
Appened one line to the greetings file.
Appened another line to the greetings file.
This is line 4
This is line 5
bodike@bodike-chaithali:~$
```

Q6)

```
This is line 9
This is line 10
This is line 11
This is line 12
This is line 13
This is line 14
This is line 15
This is line 16
This is line 17
This is line 18
This is line 19
This is line 20
This is line 21
This is line 22
This is line 23
bodike@bodike-chaithali:~$ head -n 5 greetings.txt
This is the line which creates file with in the line.
Appened one line to the greetings file.
Appened another line to the greetings file.
This is line 4
This is line 5
bodike@bodike-chaithali:~$ tail -n 15 greetings.txt
This is line 9
This is line 10
This is line 11
This is line 12
This is line 13
This is line 14
This is line 15
This is line 16
This is line 17
This is line 18
This is line 19
This is line 20
This is line 21
This is line 22
This is line 23
bodike@bodike-chaithali:~$
```